

Education

- **Purdue University** 2025–Summer 2027 (expected)
Ph.D. in Physics Advisor: Claudio Chamon.
Current research topics: condensed matter theory, quantum field theory, topological order, superconducting networks, meta-materials.
- **Boston University** 2021–2025
M.A. in Physics. Ph.D. studies in Physics; transferred to Purdue University.
- **State University of Londrina** 2019–2021
B.Sc. and M.Sc. in Physics. M.Sc. thesis advisor: Pedro R. S. Gomes.
Thesis: “Effective Field Theories for Quantum Spin Liquids and Fractonic Systems.”

Awards and Honors

- **Poster Award.** Honorable Mention for the poster “*Spatially Modulated Symmetries and Dualities*” at the 2024
Young Researchers School on Topological Aspects of Low-Dimensional Quantum Physics, Maynooth University, Ireland.
- **Goldhaber Award.** For research excellence in the first year of graduate studies at Boston University. 2023
- **Merit Award.** Ranked first among Physics undergraduate students at the State University of Londrina, with 2019
a cumulative average of 9.32/10.0.

Publications

- [11] [Guilherme Delfino](#), Mehmet Dede, Dmitry Green, Michael J. Manfra, Charles M. Marcus, and Claudio Chamon, *Kitaev spin liquid in superconducting networks*, [arXiv:2609.25210](#) (2026), preprint.
- [10] Muqing Yu, Han Bi, Hengli Lo, Vishvesha Sridhar, [Guilherme Delfino](#), Dmitry Green, Claudio Chamon, Nadya Mason, and Andrew P. Higginbotham, *Nonplanar qubit with tunable gauge symmetry*, [arXiv:2607.14229](#) (2026), preprint.
- [9] [Guilherme Delfino](#), Mehmet Dede, André L. G. Mudry, Junseok Oh, Andrew P. Higginbotham, Christopher Mudry, and Claudio Chamon, *Graph-based emulation of d-dimensional curved spaces with superconducting arrays*, [arXiv:2605.28942](#) (2026), preprint.
- [8] [Guilherme Delfino](#), Claudio Chamon, and Yizhi You, *Topological order and fractons from gauging exponential symmetries*, [Physical Review B](#) **113**, 045130 (2026).
- [7] [Guilherme Delfino](#), Dmitry Green, Saulius Vaitiekėnas, Charles M. Marcus, and Claudio Chamon, *Cooper-pair splitters as circuit elements for realizing topological superconductors*, [PRX Quantum](#) **6**, 030318 (2025).
- [6] Salvatore D. Pace, [Guilherme Delfino](#), Ho Tat Lam, and Ömer M. Aksoy, *Gauging modulated symmetries: Kramers–Wannier dualities and non-invertible reflections*, [SciPost Physics](#) **18**(1), 021 (2025).
- [5] Heitor Casasola, [Guilherme Delfino](#), Yizhi You, Paula F. Bienzobaz, and Pedro R. S. Gomes, *Fractal subsystem symmetries, anomalies, boundaries, and effective field theory*, [Physical Review B](#) **110**, 195105 (2024).
- [4] [Guilherme Delfino](#) and Yizhi You, *Anyon condensation web and multipartite entanglement in 2D modulated gauge theories*, [Physical Review B](#) **109**, 205146 (2024).
- [3] Heitor Casasola, [Guilherme Delfino](#), Pedro R. S. Gomes, and Paula F. Bienzobaz, *Fractal subsystem symmetries, ’t Hooft anomalies, and UV/IR mixing*, [Physical Review B](#) **109**, 075164 (2024).

- [2] Kai-Hsin Wu, Alexey Khudorozhkov, [Guilherme Delfino](#), Dmitry Green, and Claudio Chamon, *$U(1)$ symmetry-enriched toric code*, [Physical Review B](#) **108**, 115159 (2023).
- [1] [Guilherme Delfino](#), Wesley Bernardino Fontana, Pedro R. S. Gomes, and Claudio Chamon, *Effective fractonic behavior in a two-dimensional exactly solvable spin liquid*, [SciPost Physics](#) **14**(1), 002 (2023).

Selected Research Presentations

• Invited talks

- Max Planck Institute for the Physics of Complex Systems, Dresden June 2026
“Probing physics in arbitrary graphs and curved spaces”
- Paul Scherrer Institute (PSI), Villigen May 2026
“Gauging Modulated Symmetries”
- University of Zurich (UZH), Zurich May 2026
“Probing physics in arbitrary graphs and curved spaces”
- UCSB NSF Quantum Foundry, remote October 2025
“Topological Meta-materials from Cooper-pair Splitters”
- Quantum Theory Seminar, Purdue University, West Lafayette October 2025
“Gauging Modulated Symmetries”
- Paths to Quantum Field Theory workshop, University of Sarajevo, Sarajevo July 2024
“Gauging Modulated Symmetries in $1+1D$ ”
- Condensed Matter Theory Kids’ Seminar, Harvard University, Cambridge February 2024
“Aspects of Spatially Modulated Symmetries”
- CMSA Seminar Series, Harvard University, Cambridge November 2023
“Modulated Gauge Theories and Fracton Behavior in $2D$ ” [\[video\]](#)
- QCOMBINE Seminar Series, University of Oslo, Oslo May 2023
“Enriching Topological Order with $U(1)$ Symmetry”
- Quantum Seminar Series, Northeastern University, Boston February 2023
“Fractonic Behavior in Two Dimensions” [\[video\]](#)
- Quantum Seminar Series, Utrecht University, remote June 2022
“A Tutorial on Fractons”

• Contributed talks

- APS March Meeting, Denver March 2026
“Kitaev Quantum Spin Liquid in Superconducting Networks”
- APS March Meeting, Anaheim March 2025
“Topological Meta-materials from Cooper-pair Splitters”
- Summer School – Symmetries and Anomalies: A Modern Intake, IHES, France June 2024
“Gauging Modulated Symmetries in $1+1D$ ”
- APS March Meeting, Minneapolis March 2024
“Anyon Condensation Web in $2D$ Modulated Topological Order”
- APS March Meeting, Las Vegas March 2023
“Effective Fractonic Behavior in a Two-Dimensional Exactly Solvable Spin Liquid”

• Poster presentations

- MQC Entanglement 2026, Purdue University, West Lafayette July 2026
“Probing Physics in Arbitrary Curved Spaces through Superconducting Arrays”
- Max Planck Institute for the Physics of Complex Systems, Dresden June 2026
“Probing Physics in Arbitrary Curved Spaces through Superconducting Arrays”
- Program on Anomalies, Topology and Quantum Information in Field Theory and Condensed Matter Physics, ICTP-SAIFR, São Paulo June 2025
“Topological Meta-materials from Cooper-pair Splitters”
- Young Researchers School on Topological Aspects of Low-Dimensional Quantum Physics, Maynooth University, Maynooth April 2024
“Gauging Discrete Modulated Symmetries”

- 3rd Condensed Matter Summer School: Dynamics and Quantum Information in Many-Body Systems, University of Minnesota Twin Cities, Minneapolis
“2D Fractons from Exponential Symmetry Gauge Theories” June 2023
- XX Particles and Fields Jorge André Swieca Summer School, Campos do Jordão
“Aspects of Quantum Field Theories in 2+1 Dimensions” February 2019

Research Experience

- **Purdue University** 2025–Present
Graduate Research Assistant. Research on strongly interacting phases of matter, funded by the U.S. Department of Energy. Advisor: Claudio Chamon.
- **Boston University** 2021–2025
Graduate Research Assistant. Research on strongly interacting phases of matter, funded by the U.S. Department of Energy. Advisor: Claudio Chamon.
- **State University of Londrina** 2019–2021
Graduate Research Assistant. Research on quantum field theories and quantum matter, including effective field theories for fractonic phases of matter and quantum spin liquids; funded by CNPq. Advisor: Pedro R. S. Gomes.
- **State University of Londrina** 2017–2018
Undergraduate Research Assistant. Hamiltonian approach to black-hole thermodynamics. Advisor: Mario César Baldiotti.

Research Visits

- **Paul Scherrer Institute (PSI), Villigen, Switzerland** Summer 2026
Research visit (one week). Host: Christopher Mudry.
- **Perimeter Institute for Theoretical Physics, Waterloo, Canada** Summer 2024
Research visit (two weeks). Host: Shiyu Zhou.
- **University of Oslo, Department of Mathematics, Oslo, Norway** Summer 2023
Research visit (two weeks). Host: Lucas Hataishi.

Teaching Experience

- **Fusion Categories and SymTFT** Spring 2026
Main instructor for graduate course, State University of Londrina.
- **Modern Physics** Fall 2024
Teaching assistant for undergraduate course PY 351, Boston University.
- **Intermediate Mechanics** Fall 2023
Teaching assistant for undergraduate course PY 408, Boston University.
- **General Physics II** Spring 2022
Teaching assistant for undergraduate course PY 212, Boston University.
- **General Physics I** Fall 2021
Teaching assistant for undergraduate course PY 211, Boston University.

Academic Services

- Referee for *SciPost Physics*; *Physical Review B*; *Physical Review Letters*; *Physical Review Research*; *Physical Review X*; and *Journal of High Energy Physics (JHEP)*.